Solution

MOTION-05

Class 09 - Science

1.

(d) Displacement

Explanation: The shortest distance measured from the initial to the final position of an object is known as the displacement. Hence, the path covered from A to C is called displacement.

2.

(b) non – uniform retardation

Explanation: In the speed-time graph shown above, speed is decreasing with time. This type of graph is represented by non – uniform retardation.

3.

(b) All of these

Explanation: The image shows the motion of the moon around the earth which moves in a circle, hence, it is called circular motion.

Although the speed of the moving object remains constant, the velocity of an object changes, hence, uniform circular motion is an accelerated motion.

The change in the velocity could be due to change in its magnitude or the direction of the motion or both.

Hence, all the given statements are correct regarding the image of the circular motion of the moon around the earth.

4.

(b) 95 Km

Explanation: The object starts its journey from O which is treated as its reference point. Let A, B, and C represents the position of the object at different instants. First, the object moves through C and B and reaches A. Then it moves back along the same path and reaches C through B.

The total path length covered by the object is OA + AC, which is 60 km + 35 km = 95 km.

5.

(c) speed of A is greater than speed of B

Explanation: In the distance-time graph shown above, the speed of object A is greater than speed of object B as the object A is covering more distance per unit time.

6.

(d) 87.5 m

Explanation: Displacement of the body is the area under velocity curve = $\frac{1}{2} \times 20 \times 5$ +$\frac{1}{2} \times 20 \times 5 - \frac{1}{2} \times 10 \times 2.5$ = 10 $\times$ 5 + 10 $\times$ 5 -5×2.5= 50 + 50 - 12.5 = 87.5 m

7.

(c) Non-uniform speed

Explanation: When an object moves with a non-uniform speed it travels unequal distances in equal intervals of time. In this case, the distance traveled is not directly proportional to time and hence the graph is a curved line.

8.

(d) Becomes zero

Explanation: The distance-time graph for an object which stops moving is a straight line parallel to the time axis. When an object becomes stationary its speed is zero.

9. **(a)** Car B is slowest

Explanation: The slope of a distance time graph determines the speed of the body, so steeper the slope greater will be the speed of the body. From the above graph we can come to a conclusion that car C has the highest speed and car B has the least speed.

10.

(d) in uniform motion

Explanation: From the given v-t graph, it is clear that the velocity of the object is not changing with time i.e., the object is in uniform motion.

11.

(d) velocity of A exceeds beyond 10 ms^{-1}

Explanation: Distance = Velocity $\times$ Time = 10 $\times$ Time

The v-t graph shown here depicts the motion of A and B such that velocity of A exceeds beyond 10ms^{-1}.

12.

(c) c

Explanation: Graph c represents uniform retardation which is a straight line inclined to a time axis and the value of velocity decreases with the increase in time. Hence, the graph c shows uniform retardation.

13.

(c) a body stops moving

Explanation: The distance-time graph for an object which stops moving is a straight line parallel to the time axis.

14. **(a)** Hexagonal path

Explanation: If an athlete has to change his direction six times while he completes one round, then he is running on the hexagonal path.

15. **(a)** Non-uniform retardation

Explanation: The image shows the decrease in acceleration in the non-uniform time. Hence, the graph represents a non-uniform retardation.

16.

(d) Car B is slowest

Explanation: The slope of a distance time graph determines the speed of the body, so steeper the slope greater will be the speed of the body. From the above graph we can come to a conclusion that car C has the highest speed and car B has the least speed.

17. **(a)** velocity = 2ms^{-1}

Explanation: The position-time graph is used in physics to describe the motion of an object over a period of time. Time, in seconds, is conventionally plotted on the x-axis and the position of the object, measured in meters, is plotted along the y-axis. The slope of the position-time graph reveals important information about the velocity of the object.

18.

(c) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Explanation:

i. The slope of a Velocity – time graph of an object moving in rectilinear motion with uniform velocity in a straight line and parallel to x-axis when velocity is taken along the y-axis and time is taken along the x-axis.
ii. Graph showing uniformly accelerated velocity then straight line sloping upwards.
iii. In the velocity-time graph of a body moving with uniform retardation is a straight line sloping downwards.
iv. If an object is travelling at a variable angular speed around a fixed axis or center point, then the motion of that object will be defined as the uniform circular motion. In this case, the object travels around a curved path and it will have some radial acceleration due to which its velocity would change direction every second.

19.

(c) (a) - (iv), (b) - (i),(c) - (iii), (d) - (ii)

Explanation:

- Uniform velocity is a constant velocity parallel to the time axis.
- Acceleration is increasing velocity w.r.t time.
- Retardation is decreasing velocity w.r.t time.
- A curved line (called a parabola) is a non-uniform motion.

20.

(b) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Explanation:

- **Non-uniform motion -** A body has a non-uniform motion if it travels unequal distances in equal intervals of time.

- **Speed from the distance-time graph-** The distance-time graph of a body can be used to calculate the speed of the body. The slope of a distance-time graph indicates speed.
- **Distance covered from the speed-time graph -** The distance traveled by a moving body in a given time can be calculated from its speed-time graph. The area enclosed by the speed-time graph with the X-axis gives the distance covered.
- **Average velocity -** If the velocity of a body is always changing, but changing at a uniform rate, then the average velocity is given by the "arithmetic mean" of the initial and final velocity for a given period of time, that is Average velocity = $\frac{\text{Initial velocity }+\text{Final velocity}}{2}$

21. **(a)** (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)
Explanation:
    i. $\mathrm{I^{st}}$ equation of motion is $v=u+at$
    where the initial velocity = u, final velocity = v, Time = t and Acceleration = a
    ii. $\mathrm{II^{nd}}$ equation of motion is, S = ut +$\frac{1}{2}$at^2
    Here Initial velocity of the object = u
    The final velocity of the object = v
    Acceleration = a
    Time = t
    Distance covered in given time = s
    iii. $\mathrm{III^{rd}}$ equation of motion is $v^2=u^2+2aS$
    iv. Retardation: Negative acceleration is called retardation. If the velocity of a body is decreasing with respect to time, the acceleration is said to be negative.
    Retardation = $-\left(\frac{v-u}{t}\right)$.

22.
(b) 1-B, 2-D, 3-A, 4-C
Explanation:
    i. The SI unit of velocity is meter per second.
    ii. The SI unit of acceleration is the metre per second square.
    iii. The SI unit of displacement is the meter.
    iv. Retardation is negative acceleration.Therefore its SI unit is metre per second squared with a negative sign.

23.
(c) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)
Explanation:
- The general formula of the quantities acceleration, speed, average velocity, and third equation of motion is given.
- The velocity of a body is defined as the rate of change of its displacement with time,$v=\frac{s}{t}$.
- Acceleration of a body is defined as the rate of change of its velocity with time,$a=\frac{v-u}{t}$.
- The average velocity is given by the "arithmetic mean" of the initial and final velocity for a given period of time, that is Average velocity,$V_{av}=\frac{v+u}{2}$.
- Third equation of motion: $v^2 = u^2 + 2aS \Rightarrow v^2 - u^2 = 2aS$.

24.
(c) 1-C, 2-B, 3-D, 4-A
Explanation:
    i. A body has a non-uniform motion if it travels unequal distances in equal intervals of time.
    ii. The distance-time graph of a body can be used to calculate the speed of the body.The slope of a distance-time graph indicates speed.
    iii. The distance travelled bya moving body in a given time can be calculated from its speed-time graph.Area enclosed by the speed-time graph with the X-axis gives the distance covered.

iv. If the velocity of a body is always changing,but changing at a uniform rate,then the average velocity is given by the "arithmetic mean" of the initial and final velocity for a given period of time,that is:
Average velocity = $\frac{\text{Finalvelocity} + \text{Initial velocity}}{2}$

25.
(b) 1-D, 2-A, 3-C, 4-B
Explanation:

i. Distance travelled per unit time is called speed. SI unit for speed is m/s. Speed = $\frac{Distance\ travelled}{Time\ take}$
ii. Displacement per unit time is known as velocity. It is nothing but speed of an object in a definite direction. Velocity = $\frac{Displacement}{Time\ taken}$
iii. Rate of change of velocity with time is known as acceleration. SI unit of acceleration is m/s^2. Acceleration = $\frac{Change\ in\ velocity}{Time\ taken}$
iv. The ratio of total distance travelled to the total time taken by the body gives its average speed. Average speed = $\frac{total\ distance\ covred}{total\ time\ taken}$

26.
(c) 1-B, 2-D, 3-A, 4-C
Explanation:

i. Speed of a body is the distance travelled by it per unit time.
ii. Velocity of a body is defined as the rate of change of its displacement with time.
iii. Acceleration of a body is defined as the rate of change of its velocity with time.
iv. The average speed of an object is the total distance traveled by the object divided by the elapsed time to cover that distance.

27.
(c) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)
Explanation: Acceleration of a body is defined as the rate of change of its velocity with time.
The velocity of a body is defined as the rate of change of its displacement with time.
The speed of a body is the distance travelled by it per unit time.
When a body moves along a circular path, then its direction of motion keeps changing continuously. Therefore, the motion along a circular path is said to be accelerated.

28.
(b) (a) - (ii), (b) - (iv),(c) - (i), (d) - (iii)
Explanation:

i. The SI unit of velocity is meter per second (m/s).
ii. The SI unit of displacement is the meter (m).
iii. The SI unit of acceleration is the metre per second squared (m/s^2).
iv. The SI unit of time is the second (s).

29. **(a)** (a) - (i), (b) - (iii),(c) - (ii), (d) - (iv)
Explanation: Three equation of motion and their formula is given.
Negative acceleration is known as retardation. Therefore, the formula of retardation is the same as acceleration with a negative sign.

30.
(d) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)
Explanation: Vector- A physical quantity having magnitude as well as direction is known as a vector quantity.
Odometer- an instrument for measuring the distance travelled by a wheeled vehicle.
Acceleration of a body - It is defined as the rate of change of its velocity with time.
Scalar- A physical quantity having only magnitude is known as a scalar quantity.

31. **(a)** (a) - (i), (b) - (iii),(c) - (ii), (d) - (iv)

Explanation:

- A body has a uniform acceleration if it travels in a straight line and its velocity increases by equal amounts in equal intervals of time.
- Odometer- An instrument for measuring the distance travelled by a wheeled vehicle.
- Energy-Energy is a scalar. If it were a vector then conservation of energy would fail during uniform circular motion in radial potential and many other situations.
- Acceleration -Acceleration is a vector quantity because it has both magnitude and direction.

32.

(c) (a) - (iv), (b) - (i),(c) - (iii), (d) - (ii)

Explanation:

i. A physical quantity having only magnitude is known as a scalar quantity. A scalar quantity has no direction.
ii. A physical quantity having magnitude as well as direction is known as a vector quantity.
iii. Artificial satellites move under uniform circular motion around the earth. An artificial satellite goes around the earth in a circular orbit.
iv. Rectilinear motion is another name for straight-line motion. This type of motion describes the movement of a particle or a body.

33.

(d) 1-D, 2-A, 3-C, 4-B

Explanation:

i. Velocity of a body is defined as the rate of change of its displacement with time,v = $\frac{s}{t}$
ii. Acceleration of a body is defined as the rate of change of its velocity with time,a = $\frac{v-u}{t}$
iii. Average velocity -If the velocity of a body is always changing,but changing at a uniform rate,then the average velocity is given by the "arithmetic mean" of the initial and final velocity for a given period of time,that is: Average velocity,V_{av} = $\frac{v+u}{2}$
iv. Third equation of motion, $v^2 - u^2 = 2as$

34.

(b) (a) - (iii), (b) - (ii),(c) - (iv), (d) - (i)

Explanation: The general definition of Uniform circular motion, Velocity, Acceleration, and Speed is given.

Acceleration of a body is defined as the rate of change of its velocity with time.

The velocity of a body is defined as the rate of change of its displacement with time.

The speed of a body is the distance travelled by it per unit time.

When a body moves along a circular path, then its direction of motion keeps changing continuously. Therefore, the motion along a circular path is said to be accelerated.

35.

(b) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Explanation:

- **Scalar-** A physical quantity having only magnitude is known as a scalar quantity. A scalar quantity has no direction.
- **Vector-** A physical quantity having magnitude as well as direction is known as a vector quantity.
- **Circular motion -** Motion about a fixed point called the centre. In other words, motion in a circle is circular motion.
- **Rectilinear motion-** Rectilinear motion is another name for straight-line motion. This type of motion describes the movement of a particle or a body.

36.

(b) Negative or zero

Explanation: The actual length of the path travelled by a moving body in a given interval of time is called the distance travelled by the body. It is a scalar quantity and its value never be zero or negative.

37.

(d) Scalar

Explanation: Those physical quantities that possess the only magnitude are called scalar quantities. Distance is a scalar quantity since the direction is not required to represent distance. It is the total length of the actual path covered by a body.

38. **(a)** Uniformly

    Explanation: The relation between velocity and time is a simple one during uniformly accelerated, straight-line motion. The longer the acceleration, the greater the change in velocity. Change in velocity is directly proportional to time when acceleration is constant.

39. **(a)** 2r

    Explanation: Distance is the actual length of the path covered.
    hence, distance in this case = $\frac{2\pi r}{2} = \pi r$
    and displacement is the shortest length of path covered which in this case is the diameter = 2r

40.

    (c) Statement A
    Explanation:
    A body traveling with uniform speed travels equal distances in equal intervals of time. Hence, both the average speed and instantaneous speed are the same.
    The average speed is the total distance travelled divided by the total time taken by the object for cover up to that distance.
    Average velocity is the total displacement of objects divided by the total time taken by the object to cover up that distance.
    So, Statement A is true and Statement B is false.

41.

    (d) Statement A is true, B is false
    Explanation: When the speed of a body changes in an irregular manner, the speed-time graph of the body is a curved line. The distance traveled by the body is given by the area between the speed-time curve and the time axis. For uniform speed graph is a straight line. For non-uniform speed graph is a curved line. For Increasing speed graph Straight line sloping upwards. For decreasing speed graph Straight line sloping downwards.
    When the velocity-time graph is plotted for an object moving with uniform acceleration, the slope of the graph is a straight line.
    So, statement A is true and B is false.

42.

    (c) Both statements A and B are true
    Explanation: Uniform circular motion is a case of accelerated motion. When a body moves in a uniform circular motion then the body has an accelerated motion. It is because of a change in velocity due to continuous change in direction of motion.
    The third equation of motion, $v^2 = u^2 + 2aS$ .In this equation of motion, time is not present.
    So, both statements are true.